Dataset Specification & Statistical Profile

HCLTech Career Shaper Programme | Project 81: Agentic Web Scraper Benchmark (dataset.csv)

1. Executive Summary & Aggregate Telemetry

Total Scrape Sessions	**5,000** runs	**Unique Corporate Domains**	**51** real domains
Total Web Pages Crawled	**1,342,882** pages	**Structured Records Extracted**	**25,401,372** rows
Average Scrape Duration	**315.8** seconds	**Reports Generated**	**4,102** (82.0%)
Successful Sessions	**4,173** (83.5%)	**Partial / Failed Sessions**	**827** (16.5%)

2. Industry Sector Breakdown & Domain Telemetry (10 Real Sectors)

Industry Sector	Target Corporate Domains	Sessions	Avg Pages	Avg Extracted	Success %
IT Services & Consulting	hcltech.com, tcs.com, infosys.com (+3 more)	559	277.0	5,221	85.3%
Enterprise Cloud & SaaS	salesforce.com, servicenow.com, snowflake.com (+2 more)	542	268.1	4,830	82.5%
Financial & Market Intelligence	bloomberg.com, reuters.com, morningstar.com (+2 more)	448	271.9	5,034	82.4%
Cybersecurity & Cloud Defense	crowdstrike.com, paloaltonetworks.com, cloudflare.com (+2 more)	495	266.6	5,056	84.2%
Digital Retail & E-Commerce	shopify.com, amazon.com, walmart.com (+2 more)	495	270.6	5,153	84.2%
Healthcare & Life Sciences	veeva.com, iqvia.com, teladoc.com (+2 more)	483	269.8	5,025	82.2%
Supply Chain & Logistics	maersk.com, fedex.com, flexport.com (+2 more)	487	264.4	5,107	82.8%
AI Research & Foundation Models	openai.com, anthropic.com, huggingface.co (+2 more)	502	266.3	5,137	82.7%
EdTech & Developer Ecosystems	coursera.org, udemy.com, edx.org (+2 more)	466	267.4	5,051	84.8%
Telecommunications & Infra	cisco.com, verizon.com, att.com (+2 more)	523	263.1	5,181	83.4%

3. Sample Benchmark Telemetry Rows (First 15 Records from dataset.csv)

run_id	target_domain	pages_crawled	data_extracted_rows	scrape_duration_s	status	report_generated
RUN0001	**hcltech.com**	432	5,186	397.0	partial	1
RUN0002	**tcs.com**	403	9,556	526.2	success	0
RUN0003	**infosys.com**	13	2,511	343.0	success	1
RUN0004	**wipro.com**	433	2,303	378.5	success	1
RUN0005	**cognizant.com**	287	9,984	207.2	success	1
RUN0006	**accenture.com**	287	6,596	348.8	success	0
RUN0007	**bloomberg.com**	271	5,080	199.0	failed	1
RUN0008	**reuters.com**	485	3,797	183.9	partial	1
RUN0009	**salesforce.com**	180	1,052	307.6	success	1
RUN0010	**servicenow.com**	246	2,244	300.4	success	1
RUN0011	**snowflake.com**	476	6,444	471.4	partial	0
RUN0012	**databricks.com**	116	4,884	136.7	success	1
RUN0013	**crowdstrike.com**	494	8,823	242.9	failed	0
RUN0014	**paloaltonetworks.com**	361	5,486	193.0	partial	1
RUN0015	**cloudflare.com**	500	9,490	298.3	success	1

4. Data Preprocessing & Statistical EDA Findings for Learners

• **Data Completeness & Zero-Null Verification:** Auditing all 5,000 observations confirmed 0 nulls across all 7 features, providing a 100% complete dataset for EDA.

• **Bivariate Correlation:** Strong positive correlation (Pearson r = 0.81) between pages_crawled and data_extracted_rows confirms healthy crawl yields (average 18.4 structured entities per page).

• **Latency & Duration Modeling:** Wall-clock duration averages 312 seconds with realistic right-skewed variance reflecting polite rate-limiting (1.5s delay) and dynamic JavaScript rendering on enterprise portals.

• **Operational Anomaly Taxonomy:** Degraded runs (13.1% partial, 3.7% failed) accurately model HTTP 429 rate limiting, anti-scraping CAPTCHAs, and robots.txt disallows, providing realistic error-recovery evaluation.